

# Equivalent Expressions

Lesson 6-6

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$2x + 4$  and  $2(x + 2)$ : when  $x = 3$ , both = 10; when  $x = 7$ , both = 18 — always the same

### Equivalent

Expressions that always have the same value.

$3x + 2x + 5$  simplifies to  $5x + 5$  — fewer terms, same value

### Simplify

To write an expression in a shorter, simpler way.

$3x$  and  $7x$  are like terms (both  $x$ );  $3x$  and  $3y$  are NOT ( $x$  vs  $y$ );  $3x$  and  $3x^2$  are NOT ( $x$  vs  $x^2$ )

### Like Terms

Terms with the same letter, like  $2x$  and  $5x$ .

$4n + 3n = 7n$  — add the coefficients ( $4 + 3 = 7$ ) and keep the variable ( $n$ )

### Combine

To add or subtract terms with the same letter.

In  $5x + 3$ , the coefficient of  $x$  is 5 — when  $x = 2$ , the  $5x$  part equals 10

### Coefficient

The number in front of a letter, like the 3 in  $3x$ .

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

**I can show that two expressions are equivalent by simplifying and combining like terms.**

#### 1 Notice

At the music studio, two producers wrote formulas for the cost of renting equipment.

#### 2 Key idea

I can show that two expressions are equivalent by simplifying and combining like terms.

#### 3 Words I'll use

Equivalent

Simplify

Like Terms

Combine

Coefficient



#### Think About It

- Why did both formulas give the same answer?
- What did Producer B do differently from Producer A?
- Would they still match if  $x = 10$ ?



#### Watch

Watch your teacher model one example. Jot what you see.

---



#### We try

Solve the next one together as a class.

---

---



#### You try

Now try one on your own.

---

---

## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**Producer A wrote  $3x + 2x + 10$  and Producer B wrote  $5x + 10$ . They look different but both give 30 when  $x = 4$ . Why are they equivalent?**

#### Level 1 support

**Start here:**  $3x + 2x$  combines to  $5x$ , so the two forms are equivalent.

 **Need a hint?**

**Sentence starters (optional)**

**These expressions are equivalent because  $3x + 2x$  \_\_\_\_.**

*Estas expresiones son equivalentes porque  $3x + 2x$  \_\_\_\_.*

**I checked by substituting  $x =$  \_\_\_\_ and getting \_\_\_\_ both ways.**

*Lo comprobé sustituyendo  $x =$  \_\_\_\_ y obteniendo \_\_\_\_ en ambas.*

**Word bank:** Equivalent Simplify Like Terms Combine

**Listen for:** Students explain  $3x + 2x = 5x$  (combining like terms), so both equal  $5x + 10$ , and verify by substituting a value like  $x = 4$  to get 30 both ways.

#### Level 2

**Is testing only  $x = 4$  enough to be sure two expressions are equivalent? Justify why or why not.**

- Testing one value is \_\_\_\_ enough because \_\_\_\_.
- To be sure they are equivalent, I need \_\_\_\_.

**EXPLORE**

**How can you prove  $2(n + 4)$  and  $2n + 8$  are equivalent? What two strategies could you use?**

**Level 1 support**

**Start here:** Distribute or substitute values to show the forms always match.



**Need a hint?**

**Sentence starters (optional)**

**They are equivalent because distributing  $2(n + 4)$  gives \_\_\_\_.**

*Son equivalentes porque distribuir  $2(n + 4)$  da \_\_\_\_.*

**I can also check by substituting  $n =$  \_\_\_\_ and getting \_\_\_\_.**

*También puedo comprobar sustituyendo  $n =$  \_\_\_\_ y obteniendo \_\_\_\_.*

**Word bank:** **Equivalent** **Simplify** **Combine** **Like Terms**

**Listen for:** A strong answer uses the distributive property ( $2(n + 4) = 2n + 8$ ) and/or substitutes a value into both to confirm equal results.

**Level 2**

**Are  $2x + 3$  and  $5x$  equivalent? Find a value of  $x$  that proves your answer and explain why one test value showing equality is not enough.**

- They are \_\_\_\_ equivalent because at  $x =$  \_\_\_\_ they give \_\_\_\_ and \_\_\_\_.
- One matching value is not proof because \_\_\_\_.

CONNECT

**Client A pays  $4h + 2h + 30$  and Client B pays  $6h + 30$ . Is the pricing fair (equivalent)? How can you prove it?**

Level 1 support

**Start here:** Combine  $4h + 2h$  to  $6h$ , showing both clients pay the same.

 **Need a hint?**

**Sentence starters (optional)**

**The expressions \_\_\_\_ equivalent because  $4h + 2h =$  \_\_\_\_.**

*Las expresiones \_\_\_\_ equivalentes porque  $4h + 2h =$  \_\_\_\_.*

**I can prove it by substituting  $h =$  \_\_\_\_: both equal \_\_\_\_.**

*Puedo probarlo sustituyendo  $h =$  \_\_\_\_: ambas dan \_\_\_\_.*

**Word bank:** Equivalent Simplify Combine Like Terms

**Listen for:** Students simplify  $4h + 2h + 30$  to  $6h + 30$ , conclude the pricing is fair/equivalent, and verify by substituting a value of  $h$ .

Level 2

**A third client pays  $5h + 30$ . Critique whether that is the same deal, and explain for which client it is cheaper as hours grow.**

- $5h + 30$  is \_\_\_\_ the same because \_\_\_\_.
- As hours grow, the \_\_\_\_ client pays less because the coefficient of  $h$  is \_\_\_\_.

## Guided Examples

### Example 1

**Which expression is equivalent to  $4x + 3x$ ?**

**Solution:**  $4x + 3x = 7x$ . Combine the coefficients:  $4 + 3 = 7$ .

**Answer:** A.  $7x$

### Example 2

**Which expression is equivalent to  $2(m + 5)$ ?**

**Solution:**  $2(m + 5) = 2m + 10$  using the distributive property.

**Answer:** A.  $2m + 10$

### Example 3

**Which property is shown:  $3(x + 7) = 3x + 21$ ?**

**Solution:** Multiplying 3 by each term inside the parentheses is the distributive property.

**Answer:** A. Distributive

## Write About the Math

### The Writing Revolution

I can explain my work using the words equivalent, simplify, like terms, and combine.

#### 1. Kernel Sentence subject + verb

**Model:** Equivalent is expressions that always have the same value.

*Equivalente es expresiones que siempre tienen el mismo valor.*

**Write a kernel sentence about equivalent. Use a subject and a verb.**

*Escribe una oración base sobre equivalente. Usa un sujeto y un verbo.*

---

---

#### 2. Sentence Expansion because · but · so

**Kernel:** Equivalent matters in math

*Equivalente importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Equivalent matters in math because \_\_\_\_.**

*Equivalente importa en matemáticas porque \_\_\_\_.*

---

**but**  
*pero*

**Equivalent matters in math, but \_\_\_\_.**

*Equivalente importa en matemáticas, pero \_\_\_\_.*

---

**so**  
*entonces*

**Equivalent matters in math, so \_\_\_\_.**

*Equivalente importa en matemáticas, entonces \_\_\_\_.*

---

### 3. Sentence Types 4 ways to write a math idea

**Statement**  
*Afirmación*

Tell one true fact about equivalent.  
*Di un hecho verdadero sobre equivalent.*

**Equivalent** \_\_\_\_.

---

**Question**  
*Pregunta*

Ask a question about equivalent.  
*Haz una pregunta sobre equivalent.*

**How does** \_\_\_\_ ?

*¿Cómo* \_\_\_\_ ?

---

**Exclamation**  
*Exclamación*

Show excitement about equivalent.  
*Muestra entusiasmo sobre equivalent.*

**Wow,** \_\_\_\_ !

*¡Guau,* \_\_\_\_ !

---

**Command**  
*Mandato*

Tell a partner what to do with equivalent.  
*Dile a un compañero qué hacer con equivalent.*

**First,** \_\_\_\_ .

*Primero,* \_\_\_\_ .

---

### 4. Explain Your Reasoning use a sentence starter

**These expressions are equivalent because  $3x + 2x$  \_\_\_\_.**

*Estas expresiones son equivalentes porque  $3x + 2x$  \_\_\_\_.*

**I checked by substituting  $x =$  \_\_\_\_ and getting \_\_\_\_ both ways.**

*Lo comprobé sustituyendo  $x =$  \_\_\_\_ y obteniendo \_\_\_\_ en ambas.*

**They are equivalent because distributing  $2(n + 4)$  gives \_\_\_\_.**

*Son equivalentes porque distribuir  $2(n + 4)$  da \_\_\_\_.*

---

---

---



## Try It

Solve on your own. Check the answer key when you are done.

**1. Which property is shown:  $3(x + 7) = 3x + 21$ ?**

- A. Distributive
- B. Commutative
- C. Associative
- D. Identity

Show your work:

---

---

---

**2. Simplify:  $5n + 2n$**

- A.  $7n$
- B.  $10n$
- C.  $7n^2$
- D.  $52n$

Show your work:

---

---

---

## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A student says  $2x + 3$  and  $5x$  are equivalent because both equal 5 when  $x = 1$ . Is the student correct? Explain why testing one value is not enough to prove equivalence, and find a value of  $x$  where they give different results.**

*Sentence starter: The student is \_\_\_\_ because \_\_\_\_\_. When  $x = 1$ :  $2(1) + 3 = \underline{\hspace{1cm}}$  and  $5(1) = \underline{\hspace{1cm}}$ . But when  $x = \underline{\hspace{1cm}}$ :  $2(\underline{\hspace{1cm}}) + 3 = \underline{\hspace{1cm}}$  and  $5(\underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$ . To prove equivalence, expressions must be equal for \_\_\_\_ values of  $x$ .*

Show your work:

---

---

---

---

## Reflect — Exit Ticket

**Which expression is equivalent to  $6n + 4 + 3n - 1$ ?**

- A.  $9n + 3$
- B.  $9n + 5$
- C.  $63n$
- D.  $18n + 4$

Your answer:

---

---

---

---

## Answer Key & Teacher Guide

1. **Try It 1:** A. Distributive — *Multiplying 3 by each term inside the parentheses is the distributive property.*
2. **Try It 2:** A.  $7n - 5n + 2n = 7n$ . *Add the coefficients:  $5 + 2 = 7$ . The variable stays as  $n$ .*
3. **Exit Ticket:** A.  $9n + 3 - 6n + 3n = 9n$  and  $4 - 1 = 3$ , *so the simplified expression is  $9n + 3$ .*

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equivalent is expressions that always have the same value.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).